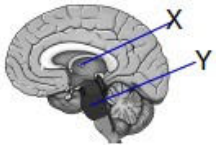
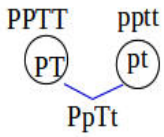
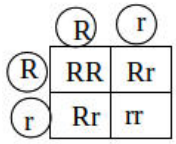
SSLC Model Examination 2026 BIOLOGY

Qn	Hints to Answers By Rasheed Odakkal, 9846626323 Kondotty	Score
1	(b). X- Phosphate Y- Sugar Z-Nitrogen base	2. (d). (i) is incorrect, (ii) and (iii) are correct.
3	(b). X- (iii), Y- (iv), Z-(ii).	4. (a). Statement is true, Reason is false.	**4x1**
5	a). Crossing over. b). Chiasma. Chromatids break at this point where chromosomes pair up.	
6	**A.** a). Pupil constricts. b). Circular muscles, which contract. **OR** **B.** When we breath, the olfactory particles enters the nasal cavity and dissolve in the mucus produced by mucus membrane. Then the olfactory receptors are stimulated and an impulse thus generated reaches the olfactory center in the brain through the olfactory nerve. Smell is perceived.	
7	a). DNA fingerprinting b). Due to differences in the arrangement of nucleotides in DNA, there are differences in DNA patterns between individuals. However, there are also similarities between parents, offspring, and siblings. These are analyzed and identified.	
8	a). X- B+ve, Y- O+ve. b). Based on the presence or absence of antigens A and Bon the surface of RBCs.	
9	The axon will lose its insulation, protection, colour, and nutrition. The speed of impulses decrease.	
10	a). X- gRNA (guideRNA), Y- Cas9 enzyme. b). The guide RNA identifies the exact DNA segment to be cut. The Cas9 enzyme cuts that section.	
11	**A.** (a). R- Filariasis. Filarial worms. (b). Female Culex mosquitoes are the vectors of the disease. Avoid circumstances to prevent mosquitoes from multiplying. **OR** **B.** a). Sickle cell anaemia. Defects in gene that control the production of haemoglobin. b). Red blood cells become sickle-shaped, reducing their oxygen-carrying capacity. These can struck in the capillaries and block blood flow.	**7x2**
12	a). X- Thalamus, Y- Pons. b). If Z (cerebellum) damaged, muscle coordination will be lost and the body will lose its balance. 	
13	**A.** a).  - Purple flowered tall pea plants. b). Purple flowered dwarf pea plants. White flowered tall pea plants. **OR B.** a). Incomplete dominance/ The dominant allele cannot fully hide the the recessive trait. (b).  1:Red flower 2: Pink flowers 1: White flower	
14	a). Thromboplastin. Formed by the platelets and damaged tissues at the wound site. b). Thrombin converts the inactive fibrinogens into fibrins.	
15	Birds develop favourable or harmful variations. When soft fruits are no longer available, there is competition for survival among birds. Only organisms with favorable variations for hard-shelled fruits survive. They reproduce more effectively and produce new generations (survival of the fittest). They persist through generations through natural selection.	
16	**A.** (a). X- Adrenal medulla, Y- Adrenal cortex. (b). They act with the sympathetic nervous system to overcome any emergency situation. (c). Aldosterone. Helps to maintain the equilibrium of minerals in the body. **OR** **B.** (a). X- Parathyroid. Parathormone. (b). If there is a low level of calcium in the blood, it helps to get it from the kidneys, intestines and bones. (c). Calcitonin. Thyroid.	**6x3**
17	(a). The prson Y. The three-month average blood glucose level of HbA1c tests should be less than 5.7 (b). If diabetes is not controlled, it can lead to heart disease, stroke, kidney disease, and vision problems. (c). Diet control, exercise, proper lifestyle...	
18	**A.** (a). X- Semicircular canals. Y- Vestibule (utricle & sacule) (b). Rotational movements of head cause the movement of endolymph in the 3 semicircular canals. Linear movements of head cause the movement of endolymph in the vestible. Both of these cause impuls production by the stimulation of hair cells there. (c). When impulses reach the cerebellum via the vestibular nerve, the brain also coordinates impulses from the eyes and muscles to correct the body's balance. **OR** **B.**(a). X- Iris, Z- Ciliary muscles. (b). Cornea becomes opaque (Xerophthalmia) due to prolonged deficiency of vitamin A. (c). Melanin. It helps to color the iris, absorb ultraviolet rays, and regulate light intensity. (d). The ciliary muscles regulate the curvature of the lens (Helps for power of accommodation).	**1x4** **(40)**